

电路与电子学实验时序 逻辑电路实验

时序逻辑电路实验

1. 实验目的:

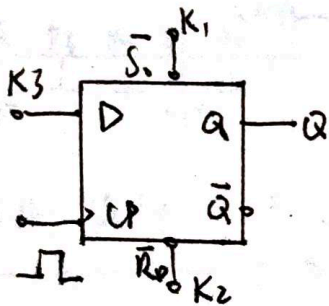
掌握集成触发器功能, 74LS161, 74LS192使用方法, 设计任意进制计数器的方法, 时序逻辑电路设计基本流程方法

2. 实验设备:

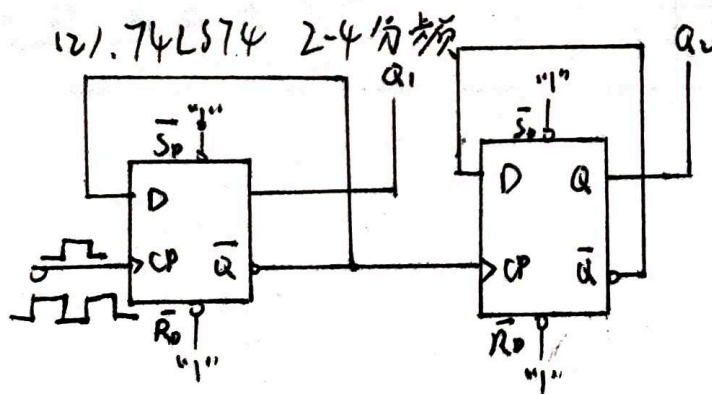
DP832电源, Agilent U124B万用表, 74LS161 74LS192 74LS74.

3. 实验内容:

(1) 74LS74基本逻辑功能验证.



CP	$\bar{R}_0$	$\bar{S}_0$	D	Q^{n+1}	$\bar{Q}^{n+1}$
X	0	1	X	0	1
X	1	0	X	1	0
X	0	0	X	1	1
↓	1	1	X	Q^n	$\bar{Q}^n$
↑	1	1	0	0	1
↑	1	1	1	1	0



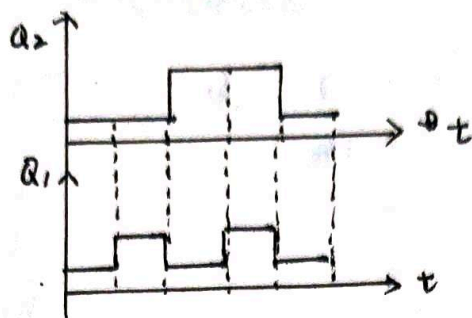
①. 用手动单脉冲作驱动信号.

用十位逻辑电平模块指示灯显示.

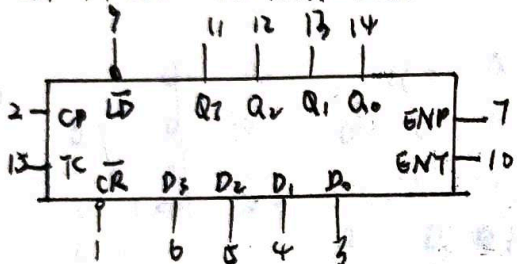
②. 用1kHz连续脉冲作驱动信号.

CP, Q1, Q2 波形

CP	Q_2	Q_1
0	0	0
1	0	1
2	1	0
3	1	1
4	0	0
5	0	1
6	1	0

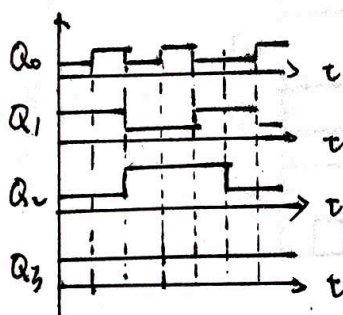
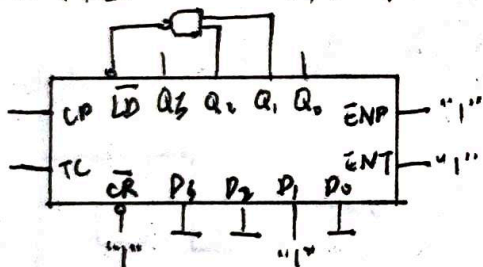


13) 74LS161 功能验证

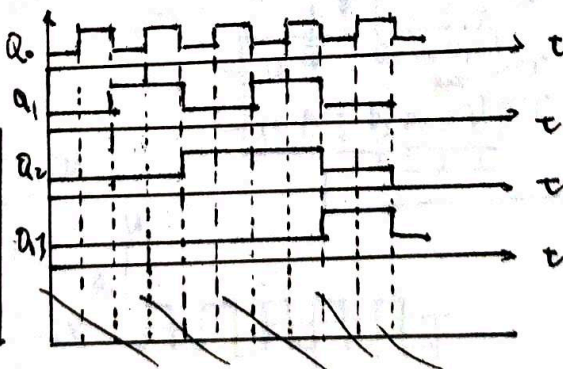
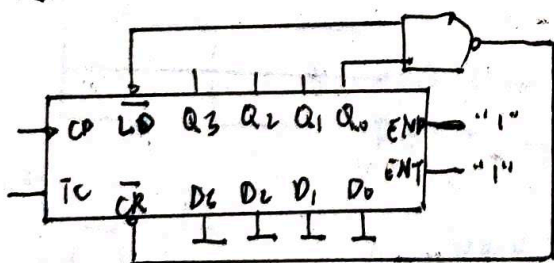


工作	输入						Q_n, Q_{n+1}
EN	CR	CP	ENP	ENT	$\bar{LD}$	D_n	Q_n
复位	0	X	X	X	X	X	0
并行in	1	↑	X	X	0	1/0	1/0
保持	1	X	0	0	1	X	保持
保持	1	X	0	1	1	X	保持
保持	1	X	1	0	1	X	保持
计数	1	↑	1	1	1	X	保持

14) 74LS161 2-6 计数



15) 74LS161 设计 0-9



时序逻辑电路实验原始数据记录

1. (1)

CP	$\bar{R}_0$	$\bar{S}_0$	D	Q^{n+1}	$\bar{Q}^{n+1}$
x	0	1	x	0	1
x	1	0	x	1	0
x	0	0	x	1	1
↓	1	1	x	Q^n	$\bar{Q}^n$
↑	1	1	0	0	1
↑	1	1	1	1	0

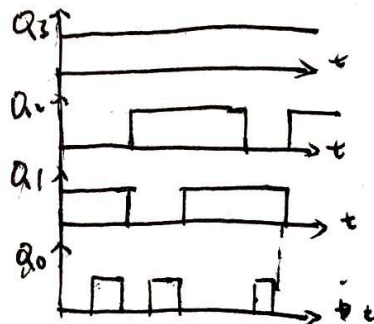
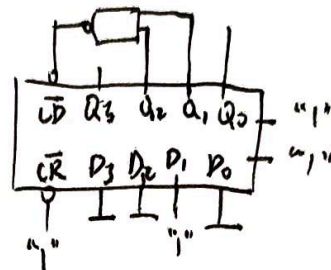
(3)

0	x	x	x	x	x	0
0	↑	x	x	0	1/0	1/0
1	x	0	0	1	x	保持
1	x	0	1	1	x	保持
1	x	1	0	1	x	
1	↑	1	1	1	x	计数

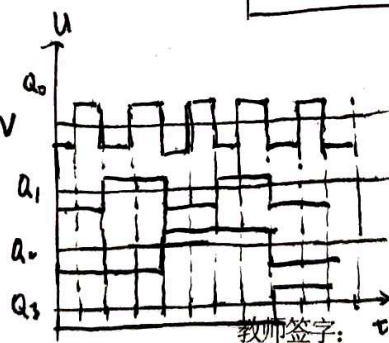
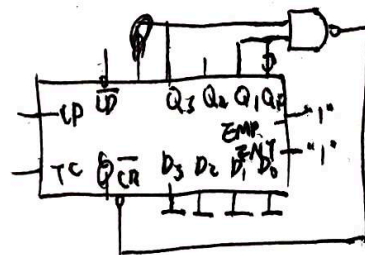
(2)

	Q_0	Q_1	Q_2	Q_3
0	0	0	0	0
1	0	1	1	0
2	1	0	0	1
3				
4				
5				
6				

(4)



(5)



教师签字: 侯云

(注: 原始数据与实验报告不允许用铅笔记录和书写)